

UNIVERSITY OF SOUTHERN CALIFORNIA

Live on version 9

Session 3: a real phenotype and a real PGI on All of Us

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"Life is not invulnerable, but it is extraordinarily good at finding ways through adversity..."

— Philip Ball, *How Life Works* (2023)

20 minutes of instruction 40 minutes of practice

What changes when the data are real

- **Nothing in the workflow.** Fetch, build, clean, look, score, save – the same commands, now against v9. That was the plan all along.
- **Everything in the stakes.** Person-level files stay in the workspace, forever. Aggregates follow the dissemination rules. Screens still share what you show.
- **Your numbers are yours.** No answer key today: results are real, and cohorts drift as data are released. The handout gives plausibility checks, not answers.
- **Queries and copies cost money** – capped in our scripts, billed to the workshop project.

Same muscle memory, real consequences. That is why we practiced.

Where today's data live

Resource	What we take from it
CDR (BigQuery)	Height measurements + person table → our phenotype
SSGAC shared bucket	HapMap3-filtered PLINK fileset, one chromosome
v9 controlled bucket	Genetic-ancestry predictions (requester-pays: fetches must name a billing project)
Workshop bucket	Instructor-staged PGI weights; your saved runs

Cloud files must be **copied to the VM disk** before PLINK can touch them. Copies are fast inside Google's network – and counted, every time.

Paths are variables in `config.aou.sh` – which is why the same scripts also run against the local sandbox.

A phenotype is a chain of decisions

```
WHERE measurement_source_concept_id = 903133    -- not LOINC 3036277
```

- **Concept choice:** the program's own measurement concept. The standard LOINC concept “works” – and silently pools in hundreds of thousands of EHR heights.
- **One row per person:** we take each person's *median* (mean and first-visit are rival, defensible choices – pick one, state it).
- **Plausibility bounds:** 100–250 cm, with removals counted.
- **Standardize within sex;** “other” handled explicitly, not silently.
- Still pooled underneath: clinic and remote self-report share this concept.

Every line of the cleaning script is a methods sentence.

Real files, real conventions

File	Its quirk
.fam (SSGAC fileset)	Key spelled FID=IID; other AoU filesets use FID=0 – a mismatched keep-list matches zero samples, silently
.bim	rsIDs here; AoU ACAF callsets use variant IDs – crossing schemes needs the VAT map
ancestry_preds.tsv	Key column named research_id – same numbers as person_id, different name
.pheno files	Can contain literal NA: a value that crashes or poisons, depending on your parser

Read two lines of every file before joining it. Count every join.

Cleaning and EDA, with disclosure built in

- First look at the phenotype: N, mean, SD, quartiles – overall and by sex. Is the mean plausible for a US cohort? Is the SD? Would a spike or second mode implicate a concept or unit upstream?
- Printed counts are **rounded to the nearest 100**, and 1–20 never appears – so no table cell, or difference of cells, can disclose a small count. Exact Ns exist only in workspace-side files.
- Histogram bars are counts too; sparse bins can whisper. Figures get a human look before any dissemination.

The funnel stays narratable and disseminable at the same time.

Real weights, real match counts

```
p2 --bfile lab_data/chr22_filtered \  
  --score lab_data/height_weights.txt 1 2 3 header \  
    cols=+scoresums list-variants --out results/aou_pgi
```

- Weights are **instructor-staged** from a public GWAS release, rsID-keyed to match this fileset. Your first read of the log is the match count.
- One chromosome $\Rightarrow$ a deliberately **partial** PGI. Real studies score all 22 and sum – same command, a loop, and patience.
- Missing genotypes are mean-imputed by default; state it.

The mechanics you rehearsed twice are unchanged. Only the files got real.

Reading the ancestry figure honestly

The lab's last figure overlays PGI densities by AoU's predicted genetic ancestry.

- Distributional differences reflect **allele frequencies and the discovery GWAS behind the weights** as much as any biology. Say so wherever the figure appears.
- **Portability**: prediction accuracy is not constant across groups – a PGI built from mostly-European discovery GWAS predicts less well at greater ancestral distance. In AoU this determines who your index serves.
- Ancestry here is a statistical prediction from genotypes – not race, not ethnicity, not a quality score.

Two facts, always together: distributions differ, and accuracy differs.

Costs, time, and etiquette

- Queries are byte-capped ($\sim$ \$0.20 each at the default); bucket fetches bill the workshop project; today's whole lab is a few dollars per seat.
- Default cohort: a **1-in-10 teaching subset** (`SAMPLE_MOD=10`) – fast queries, quick joins. `SAMPLE_MOD=1` is the same code on everyone, for your real projects.
- The sandbox rehearsed everything except BigQuery before class – that is why the commands will feel familiar. A green sandbox run still proves plumbing, not science.
- **Stop your cloud app when done.** VMs bill while idle.

Cheap, capped, rehearsed – and turned off afterward.

Your 40-minute lab

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- 20–26 Copy `config.aou.example.sh` → `config.aou.sh`; fill values; pre-flight.
 - 26–32 Stage the chromosome + ancestry file; build the phenotype from the CDR.
 - 32–38 Phenotype EDA: read the funnel, open both figures.
 - 38–48 Fetch weights, score, PGI EDA; open the ancestry figure.
 - 48–54 Interpretation questions with a partner.
 - 54–60 Save (the gate is live), **stop the app**, exit check.
-

```
cp config.aou.example.sh config.aou.sh    # then edit its values
export WORKSHOP_CONFIG=config.aou.sh
bash scripts/00_preflight.sh
bash scripts/10_fetch_aou_genotypes.sh && bash scripts/11_build_phenotype.sh
bash scripts/12_pheno_eda.sh && bash scripts/13_score_real_pgi.sh
bash scripts/04_save.sh
```

Questions to answer while your code runs

1. Which allele-and-ID conventions did today's three joins have to survive?
2. Why the median per person? Name one alternative and its trade-off.
3. Your funnel prints ~59,300. Why not the exact number?
4. What match rate did `-score` report, and what does a low one mean?
5. Give the two-sentence honest reading of the ancestry figure.
6. What would change in the code to run everyone instead of 1-in-10? To score all 22 chromosomes?

The exit check

“I pulled real records, defended every cleaning step, counted every join, and read a real PGI honestly.”

For your own project

Swap the concept ID, the bounds, and the one-row rule for your phenotype – the chain is the same.

Swap the staged weights for your trait's release, and score all 22 chromosomes.

Keep the defaults that protected you today: caps, counts, rounding, the gate, and the sandbox first.

“... the astonishing process that created a form of matter able to begin understanding itself: us.”

— Philip Ball, *How Life Works* (2023)

Jeff Swigert USC CESR All of Us live-data workshop